Veille informatique quantique - 2026-10-02

Période analysée : les 7 derniers jours. Articles retenus : 10.

Synthèse

1. A Gate-Based Quantum Computing Framework for Codon Optimization

Source : arXiv Quantum Physics - publié le 2026-10-02

Scores : pertinence 87/100 | nouveauté 50/100 | importance 50/100 | score 67/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.00887v1 Announce Type: new Abstract: Codon optimization is a challenging combinatorial optimization problem with important applications in synthetic biology, protein expression, and biotechnology. While quantum annealing has previously been explored for this problem, the application of gate-based quantum algorithms remains largely unexplored. In this work, we present a gate-based quantum computing framework for codon optimization by formulating the optimization objective as an Ising

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computing, quantum processor, quantum algorithm, IBM Quantum.

2. Less precise but less noisy: local circuits for momentum-space state preparation and measurement

Source : arXiv Quantum Physics - publié le 2026-10-02

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.01704v1 Announce Type: new Abstract: Quantum algorithms are usually optimized for gate count or circuit depth. We find on Quantinuum System Model H2 quantum computer that for a tight-binding chain ground state preparation, there is a system size N beyond which the adiabatic evolution reaches significantly lower energies than the Fermionic Fourier Transform (FFT), with the same number of gates, and with the same circuit depth. We attribute this high noise sensitivity of the FFT to i

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Mots-clés détectés : quantum computer, quantum algorithm, Quantinuum.

3. Adaptivity is all you need: Optimal stabilizer learning using just single-copy measurements

Source : arXiv Quantum Physics - publié le 2026-10-02

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.02031v1 Announce Type: new Abstract: Stabilizer states are central to quantum computing, underlying quantum error correction, benchmarking, and efficient classical simulation. Yet their learnability exhibits a striking gap: an n-qubit stabilizer state can be learned from $\Theta(n)$ copies using two-copy Bell measurements, whereas non-adaptive single-copy measurements require $\Omega(n^2)$ copies. Here we show that adaptivity completely closes this gap. We give a polynomial-time ad

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computing, qubit, quantum error correction.

4. Spiking neural networks for streaming qubit readout

Source : arXiv Quantum Physics - publié le 2026-10-02

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.02129v1 Announce Type: new Abstract: Fast and accurate qubit-state assignment is essential for feedback, calibration, and error correction in quantum processors. In superconducting platforms, frequency-multiplexed readout makes this task intrinsically multivariate as measured traces can encode crosstalk, qubit-state relaxation events, and other transient nonidealities that are not fully captured by conventional matched filtering. Here, we introduce spiking neural network (SNN) discri

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : qubit, superconducting qubit, quantum processor.

5. Coherence Rather Than Error Rate Governs Privacy in Multi-Tenant Quantum Computing

Source : arXiv Quantum Physics - publié le 2026-10-02

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.34411v2 Announce Type: replace Abstract: Multi-tenant computing enables providers of commercial cloud quantum processors to rent disjoint sectors of a device to independent users. Average gate error, which cloud quantum computing providers report, does not determine how much one tenant learns about another. We propose an information-theoretic notion of information leakage across co-tenancy boundaries stemming from quantum state distinguishability. We measure this leakage on commercia

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computing, qubit, quantum processor.

6. Quantum Circuit Pruning: From NISQ Architectures to Fault-Tolerant Operations

Source : arXiv Quantum Physics - publié le 2026-10-02

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.00669v1 Announce Type: new Abstract: We present a routing-aware pruning strategy for quantum circuits that selectively removes parametric two-qubit gates whose computational contribution is outweighed by the fidelity cost of their execution, and assess it across both Noisy Intermediate-Scale Quantum (NISQ) and fault-tolerant quantum computing (FTQC) architectures. Each two-qubit gate of a circuit is evaluated by comparing an architecture-independent lower bound on the fidelity cost o

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Mots-clés détectés : quantum computing, qubit.

7. Real-Time Adaptive Filtering and the Boxcar Limit in Superconducting Qubit Readout

Source : arXiv Quantum Physics - publié le 2026-10-02

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.00783v1 Announce Type: new Abstract: In dispersive readout of superconducting qubits, the common baseline is a boxcar averager: a uniformly weighted average over a fixed window followed by threshold-based state assignment. We ask when additional digital processing improves fidelity. We evaluated an adaptive finite-impulse-response (FIR) filter trained in real time by the least-mean-squares (LMS) algorithm on an FPGA quantum controller, and analyzed measured readout shots offline to c

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : qubit, superconducting qubit.

8. TATVA: A Reinforcement Learning Framework for Quantum Circuit Synthesis

Source : arXiv Quantum Physics - publié le 2026-10-02

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.00945v1 Announce Type: new Abstract: One of the most important initial steps in quantum computing is high-fidelity quantum state preparation, because errors in it can affect the final computational result. Therefore, one of the major challenges is to design a quantum circuit that can achieve the desired target state accurately with a smaller number of gates. Traditionally, circuits are designed using predefined rules and mathematical decompositions, but this makes it difficult to des

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Mots-clés détectés : quantum computing, qubit.

9. Constant-Per-Layer-Depth MPS-Pretrained Ansatz for Noisy Distributed Quantum Processors

Source : arXiv Quantum Physics - publié le 2026-10-02

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.01099v1 Announce Type: new Abstract: Distributed quantum processors could scale variational algorithms beyond single devices, but circuit depth, communication overhead, and noise limit their performance. We compare ladder, mixed-canonical, and brick-wall realizations of matrix-product-state (MPS) pretraining with matched per-layer two-qubit-block resources. Despite comparable ideal variational quantum eigensolver performance and gradient scales, the shallower brick-wall architecture

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Mots-clés détectés : qubit, quantum processor.

10. Trapdoored Clifford Operators and Applications

Source : arXiv Quantum Physics - publié le 2026-10-02

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.01848v1 Announce Type: new Abstract: Random Clifford operators have numerous applications in quantum computing, including randomized benchmarking, classical shadows, and quantum authentication. However, sampling and implementing uniformly random n-qubit Clifford incur near-quadratic complexity due to the size of Clifford group. We introduce a cryptographic way to overcome these barriers: trapdoored Clifford operator distributions whose samples are computationally indistinguishabl

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Mots-clés détectés : quantum computing, qubit.

Méthode

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